

Assignment 5

1. [Previous Final Exam Question]

Suppose TS1 to TS 4 assumptions are satisfied but the errors in a simple regression model, $y_t = \beta_0 + \beta_1 x_t + u_t$, have an $AR(1)$ serial correlation (i.e., $u_t = \rho u_{t-1} + e_t$ and e_t is serially uncorrelated with e_s where $s \neq t$). Show that the OLS standard error for $\hat{\beta}_1$ often tends to be smaller than the correct standard error.

[Hint: In one of the practice questions, we had derived the conventional variance (or standard error) $Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$ for a simple cross-sectional linear regression model, i.e., $y_i = \beta_0 + \beta_1 x_i + u_i$, $i = 1, 2, \dots, n$. In a simple time series linear regression model, the conventional the conventional variance (or standard error) $Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{t=1}^T (x_t - \bar{x})^2}$ is same as what we had learned in Econometrics I or HE2005, but we change the subscript i to t to show that we are dealing with the time series observations. Please show that this conventional formula is wrong when TS 5: No Serial Correlation Assumption is not satisfied. In addition, show that this wrong formula usually underestimates the true variance and usually provides a numerical answer smaller than the answer based on the correct variance estimate that accounts for serial correlation.]

[Difficulty level: 4 out of 5]

2. [Previous Final Exam Question]

Suppose $y_t = a_0 + a_1 y_{t-1} + u_t$, and $u_t = \rho u_{t-1} + e_t$ and e_t is serially uncorrelated with e_s where $s \neq t$.

- (a) Show that the contemporaneous exogeneity assumption does not hold
- (b) Show that the problem in (a) is solved when y_{t-2} is included in the model

[Difficulty level: 2.5 out of 5]